Pre-lab
Lab 6: Energy/Momentum Conservation

Name : ________________________

NetID : ________________________ **Grade:_____________/30**

Section :

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❖ You are encouraged to have discussions with TA and other students, but you are required to do calculations and answer questions individually and independently.

Question: [30]

(1)

i. Fill in the predicted velocity with sign and % difference between the predicted and measured values. (Express answers to 3 sig. figs.) [8]

Collision type	predicted $v_{1,f}$	% diff in $v_{1,f}$	predicted $v_{2,f}$	% diff in $v_{2,f}$
Elastic				
Completely inelastic				

ii. Fill in the initial value, final value, and % loss. (Express answers to 3 sig. figs.) [6]

Collision type	Momentum $(\mathrm{kg} \times \mathrm{m/s})$			Kinetic Energy $(\mathrm{kg} \times \mathrm{m}^2/\mathrm{s}^2)$		
	Initial $p_{1,i} + p_{2,i}$	Final $p_{1,f} + p_{2,f}$	% loss	Initial $K_{1,i} + K_{2,i}$	Final $K_{1,f} + K_{2,f}$	% loss
Elastic						
Completely inelastic						

(2) Explain the four statements about the elastic collision experiment. [8]

Answer:

(3) Show $\frac{|K_{f,1}+K_{f,2}-K_{i,1}-K_{i,2}|}{|K_{i,1}+K_{i,2}|}=\frac{m_2}{m_1+m_2}$ for a completely inelastic collision and explain how this relation tells that the energy is not conserved. [4]

Answer:

(4) Show that $v_{1,f} = v_{1,i}$ and $v_{2,f} = v_{2,i}$ satisfy the conservation equations and explain why it is not a solution to an elastic collision problem. [4]

Answer: